

Financial valuation of actuarial liabilities
Homework 1 - State bonds

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Question 1

The constant maturity rates (or par yields) for 12/01/2026 retrieved from the Banque de France website are summarized in the following table.

Maturity (years)	Constant maturity rate
1	2,13%
2	2,24%
3	2,42%
5	2,79%
7	3,12%
10	3,55%
15	3,95%
20	4,18%
25	4,36%
30	4,46%

Table 1: Constant maturity rates/ par yields on 12/01/2026.

For intermediate values, we fit a Nelson-Siegel model

$$yield(t) = b_0 + \frac{1}{t} \frac{b_{10}}{c_1} (1 - e^{-c_1 t}) + \frac{1}{t} \frac{b_{11}}{(c_1)^2} (1 - (c_1 t + 1)e^{-c_1 t})$$

characterized by 4 parameters: b_0 , b_{10} , b_{11} and c_1 . The latter must be positive, as it is the decay factor.

The optimal parameters are found by minimizing the root mean square error (RMSE) between real and model yields. Setting the RMSE as the value to minimize on the Excel solver, and choosing b_0 , b_{10} , b_{11} and c_1 as the values to change, the iterative solver converges to the following values:

Parameter	Value
b_0	0.016841372
b_{10}	0.001546384
b_{11}	0.004745281
c_1	0.052446037

Table 2: Nelson-Siegel model parameters

We note that since $c_1 > 0$, it is a valid solution. With these values, the RMSE is 0,000374604. Figure 1 compares the real yields with the ones given by the fitted Nelson-Siegel model.

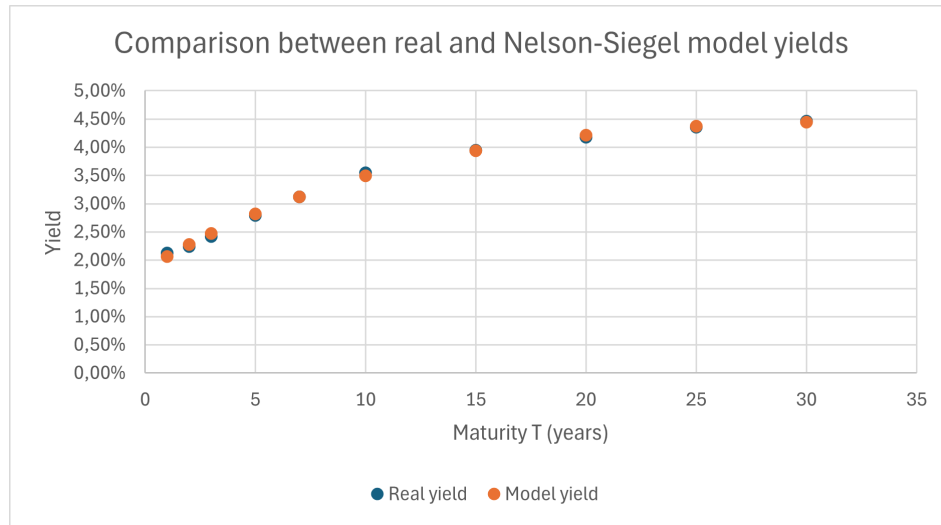


Figure 1: Comparison between real and Nelson-Siegel model yields

The tiny observed gaps between real and model yields confirm the soundness of our parameter values.

Question 2

Thanks to the Nelson-Siegel model fitted in Question 1, the intermediate values of the constant maturity rate can be estimated. So, from now on, we use the same rates retrieved from Banque de France for maturities 1, 2, 3, 5, 7, 10, 15, 20, 25 and 30 years, and take the values given by the model $yield(t)$ for other maturities. The next table lists the par rates on 12/01/2026 that will be used.

Maturity t (years)	Constant maturity rate c_t	Bootstrapped spot rate $s(t)$
1	2.13%	2.1300%
2	2.24%	2.2412%
3	2.42%	2.4253%
4	2.6498%	2.6636%
5	2.79%	2.8097%
6	2.9742%	3.0063%
7	3.12%	3.1636%
8	3.2526%	3.3089%
9	3.3762%	3.4470%
10	3.55%	3.6492%
11	3.5952%	3.6961%
12	3.6917%	3.8104%
13	3.7803%	3.9174%
14	3.8613%	4.0173%
15	3.95%	4.1311%
16	4.0029%	4.1962%
17	4.0641%	4.2764%
18	4.1196%	4.3503%
19	4.1695%	4.4181%
20	4.18%	4.4229%
21	4.2545%	4.5384%
22	4.2901%	4.5888%
23	4.3215%	4.6335%
24	4.3489%	4.6727%
25	4.36%	4.6815%
26	4.3931%	4.7362%
27	4.4102%	4.7595%
28	4.4243%	4.7776%
29	4.4356%	4.7908%
30	4.46%	4.8357%

Table 3: Constant maturity rates/par yields and bootstrapped zero-coupon/spot rates on 12/01/2026. Blue: given; Red: Nelson-Siegel approximation.

In order to bootstrap zero-coupon (or spot) rates from these par rates, let us recall that the t -year par rate is the coupon rate c_t at which a bond paying regular coupons would trade at par value, N , and that the t -year spot rate $s(t)$ is the interest rate for a zero-coupon bond maturing at time t . Therefore, bootstrapping essentially consists in determining the spot rate

value $s(t)$ which gives to the bond a value equal to its principal.

- 1-year spot rate:

$$N = \frac{(c_1 + 1)N}{1 + s(1)} \iff s(1) = c_1 + 1 - 1 = c_1$$

- 2-year spot rate:

$$N = \frac{c_2 N}{1 + s(1)} + \frac{(c_2 + 1)N}{(1 + s(2))^2} \iff s(2) = \left(\frac{c_2 + 1}{1 - \frac{c_2}{1 + s(1)}} \right)^{\frac{1}{2}} - 1$$

- 3-year spot rate:

$$N = \frac{c_3 N}{1 + s(1)} + \frac{c_3 N}{(1 + s(2))^2} + \frac{(c_3 + 1)N}{(1 + s(3))^3} \iff s(3) = \left(\frac{c_3 + 1}{1 - \frac{c_3}{1 + s(1)} - \frac{c_3}{(1 + s(2))^2}} \right)^{\frac{1}{3}} - 1$$

⋮ (Generalizing)

- n-year spot rate:

$$N = \sum_{t=1}^{n-1} \frac{c_t N}{(1 + s(t))^t} + \frac{(c_n + 1)N}{(1 + s(n))^n} \iff s(n) = \left(\frac{c_n + 1}{1 - \sum_{t=1}^{n-1} \frac{c_t}{(1 + s(t))^t}} \right)^{\frac{1}{n}} - 1$$

With this formula, it is possible to successively derive $s(1)$, $s(2)$, $s(3)$, etc. all the way to $s(30)$. Let us note that the computation of $s(t)$ requires the computation of all previous spot rates $s(t - 1)$, $s(t - 2)$, etc. This process is easily automated using Excel. One just needs to initialize the first spot rate $s(1) = c_1$. The last column of Table 3 shows the bootstrapped spot rates, and Figure 2 offers a comparison of the par rate curve with the bootstrapped zero-coupon rate curve.

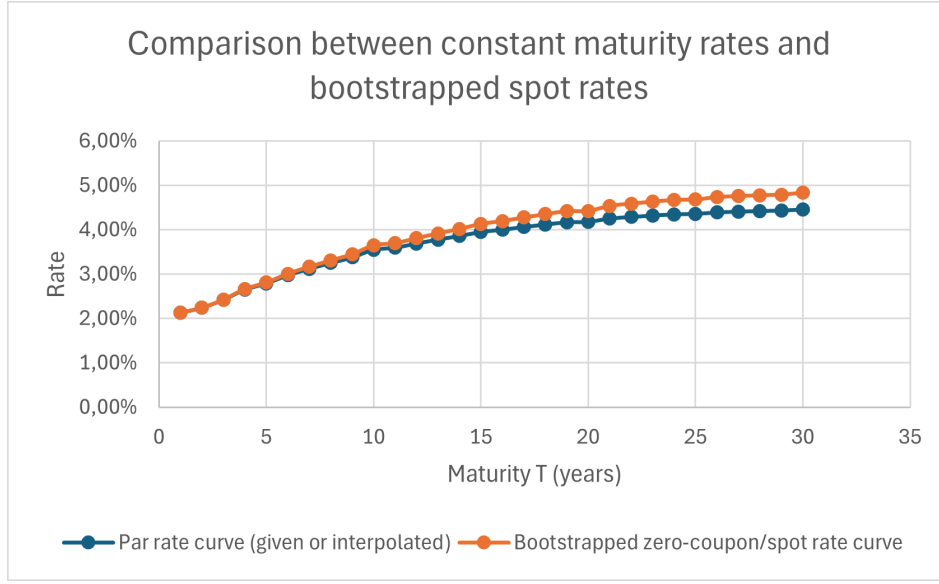


Figure 2: Comparison between constant maturity rates and bootstrapped spot rates.

We can observe that the spot rates are always slightly above the corresponding par rates, which is a normal behaviour for such a 'traditional' concave upward yield curve. Indeed, we can intuitively see par rates as weighted average of current and previous spot rates. Since the curve is concave and increasing with maturity, the long-term spot rate reflects the highest rate, and therefore the par rate, which averages lower earlier spot rates, must be smaller than the corresponding spot rate.

Using these spot rates, we can now compute the best estimate value of liabilities. The `DATAV2.xlsx` file contains premiums Π_t and benefits B_t for each year t . The best estimate being the sum of discounted net cash flows, it is calculated as follows:

$$BE = \sum_{t=1}^{30} \frac{CF_t}{(1 + s(t))^t} \quad (1)$$

where $CF_t = B_t - \Pi_t$.

Using the values from the file yields a best estimate of $BE = 1641132.554$.

Question 3

Yield of liabilities

The yield of liabilities is the anticipated total return on the investment consisting in paying the value BE to receive future cash flows $\{CF_t\}_{t=1}^{30}$. It is the value y_L such that

$$BE = \sum_{t=1}^{30} \frac{CF_t}{(1 + y_L)^t} .$$

In Excel, we can directly use the `IRR(values, [guess])` function. It takes as input the set of upcoming cash flows (the first being the payment at $t = 0$ of BE , hence negative), as well as an optional guess starting value, and outputs the yield y_L .

Using the computed cash flows CF_i and the best estimate BE , we obtain the yield $y_L = \text{IRR}(\{-BE, CF_1, CF_2, \dots, CF_30\}, 0.04) = 3.7636\%$.

Modified duration

The modified duration measures the percentage change in liability value for a 1% change in yield value y_L . It can be computed as

$$D_{mod} = \frac{D}{1 + y_L}$$

where y_L denotes the previously computed yield of liabilities and D , the Mac-Aulay duration, is the average lifetime of liabilities, that is the average time until the liabilities' cash flows are paid, with the weights equal to the present values of those cash flows:

$$D = \frac{1}{BE} \sum_{t=1}^{30} t \frac{CF_t}{(1 + y_L)^t} .$$

In our case, we obtain $D = 8.827116815$ (years) and $D_{mod} = 8.506946916$.

Note: In Excel, these computations are eased with the use of the `SUMPRODUCT(array1, [array2], [array3], ...)` function.

Convexity

The convexity formula is the following:

$$C = \frac{1}{BE (1 + y_L)^2} \sum_{t=1}^{30} t(t+1) \frac{CF_t}{(1 + y_L)^t} .$$

Direct computation of this quantity gives $C = 118.9981821$.

Question 4

Let Δ_{y_L} be the value of parallel yield shift and $BE(y_L)$ be the value of the best estimate when the yield of liabilities is y_L . Let us analyze the impact on the best estimate BE of a yield shift.

Approximation using the modified duration

The relative change of the best estimate is

$$\frac{BE(y_L + \Delta_{y_L}) - BE(y_L)}{BE(y_L)} \approx \frac{1}{BE(y_L)} \Delta_{y_L} \frac{dBE(y_L)}{dy_L} = -D_{mod} \Delta_{y_L} .$$

So for $\Delta_{y_L} = 1\%$, the relative change is -8.5069% (the BE decreases from 1641132.554 to 1501522.279) while for $\Delta_{y_L} = -1\%$, the relative change is the opposite, 8.5069% (the BE increases from 1641132.554 to 1780742.83).

Approximation using the modified duration and the convexity

The relative change of the best estimate is

$$\frac{BE(y_L + \Delta_{y_L}) - BE(y_L)}{BE(y_L)} \approx \frac{1}{BE(y_L)} \left[\Delta_{y_L} \frac{dBE(y_L)}{dy_L} + \frac{1}{2} (\Delta_{y_L})^2 \frac{d^2 BE(y_L)}{dy_L^2} \right] = -D_{mod} \Delta_{y_L} + \frac{1}{2} C (\Delta_{y_L})^2 .$$

For $\Delta_{y_L} = 1\%$, the relative change is -7.9120% (the BE decreases from 1641132.554 to 1511286.869) while for $\Delta_{y_L} = -1\%$, the relative change is 9.1019% (the BE increases from 1641132.554 to 1790507.419).

Exact revaluation

We can compute the exact values for $BE(y_L + \Delta_{y_L})$ using a slightly different version of Equation (1), taking into account the parallel shift for all spot rate values:

$$BE = \sum_{t=1}^{30} \frac{CF_t}{(1 + s(t) + \Delta_{y_L})^t} .$$

For $\Delta_{y_L} = 1\%$, the BE decreases from 1641132.554 to 1516028.949 (relative change of -7.6230%) while for $\Delta_{y_L} = -1\%$, the BE increases from 1641132.554 to 1784017.715 (relative change of 8.7065%).

For better comparison, the values are reported in Table 4.

Method	Δ_{y_L}	New BE	Absolute change	Relative change
Initial Value	–	1641132.554	–	–
Modified duration	+1%	1501522.279	-139610.275	-8.5069%
Modified duration	-1%	1780742.830	+139610.276	+8.5069%
Modified duration + Convexity	+1%	1511286.869	-129845.685	-7.9120%
Modified duration + Convexity	-1%	1790507.419	+149374.865	+9.1019%
Exact Revaluation	+1%	1516028.949	-125103.605	-7.6230%
Exact Revaluation	-1%	1784017.715	+142885.161	+8.7065%

Table 4: Impact of a $\pm 1\%$ parallel shift in yield y_L on the best estimate BE .

We observe that the duration-only estimation underestimates price increase when rates fall, and overestimates price decrease when rates rise. The duration + convexity approximation is much closer to the exact relative change, showing that convexity is a metric that matters for large shocks. In our case, the high values of durations and convexity warns us that the insurance liabilities are very sensitive to interest rate movements.

Question 5

We here delve into the 0L02.75%22APR2039 government bond. From the provided PDF file, we understand that this instrument is a Belgian government bond ("OLO" for Obligation Linéaire/Lineaire Obligatie) issued by the Kingdom of Belgium. It is a fixed-rate sovereign bond denominated in euros and traded on Euronext Brussels. The bond pays a fixed annual coupon of 2.75% of its nominal/face value. Assuming a nominal value of 100, the annual coupon payment amounts to 2.75 EUR. The bond matures on 22 April 2039, at which date the nominal value is repaid.

As for the market price information, we see that on 09/01/2026, the bond is quoted at 90.55% of its nominal value. Since this is below par (100), the bond is trading at a discount. This is consistent with the fact that current market interest rates are higher than the coupon rate of 2.75

The PDF also provides us with historical price performance and intraday trading information: a table shows the daily price information (open, high, low, close), as well as performance indicators (1W, 1M, YTD, 52W).

We are now interested in computing the accrued interest of the bond on 09/01/2026. Let us remind that the accrued interest represents the fraction of the coupon that has been earned by the seller since the last coupon payment date. Since bonds are quoted clean of accrued interest, the buyer must compensate the seller for the interest already earned and pay accrued interest in addition to the quoted clean price.

The formula to compute accrued interests is

$$AI = C \cdot DCF$$

where C is the annual coupon and DCF is the so-called day count fraction, that is the fraction of a year that represents the time between two dates, namely between today's date and the last coupon date. The DCF is calculated according to a specific day count convention that specifies how the number of days between two dates is calculated. For instance, under the "real/real" (or "actual/actual", also abbreviated "ACT/ACT") convention, we use the actual number of calendar days between dates, divided by the actual number of days in the coupon period.

Using the settlement date 09/01/2026, the last coupon date 22/04/2025 and the next coupon date 22/04/2026, we can calculate the accrued interests:

$$\begin{aligned} AI &= C \cdot \frac{\text{Number of actual days between 22/04/2025 and 09/01/2026}}{\text{Number of actual days between 22/04/2025 and 22/04/2026}} = 2.75 \cdot \frac{262}{365} \\ &= 1.973973 \text{ EUR} \end{aligned}$$

In Excel, we directly use the `ACCRINT(issue, first_interest, settlement, rate, par, frequency, [basis], [calc_method])` function. `issue` is the bond issue date which we set here to the last coupon date, `first_interest` is the date of the first coupon payment from today, `settlement` is the settlement date, or the date at which we want to evaluate the security. We set it to 09/01/2026. `rate` is the annual coupon rate (2.75%), `par` is the nominal value (100), `frequency` is the number of coupon payments per year (set to 1 since payments are annual) and `basis` is the day count convention, set to 1 to use the "ACT/ACT"

convention. This function outputs the same value:

$$AI = \text{ACCRINT}(\text{DATE}(2025;4;22); \text{DATE}(2026;4;22); \text{DATE}(2026;1;9); 2.75\%; 100; 1; 1) \\ = 1.973973 \text{ EUR}$$

Finally, the dirty price is the total amount paid by the buyer at settlement, including compensation for accrued interests:

$$\text{Dirty price} = \text{Clean price} + AI = 90.55 + 1.973973 = 92.523973 \text{ EUR}$$

Question 6

The yield of the bond is the constant interest rate y_B such that the sum of discounted cash flows to the date 09/01/2026 is equal to the dirty price:

$$\text{Dirty price} = \sum_{t=1}^{14} \frac{2.75}{(1 + y_B)^{t-DCF}} + \frac{100}{(1 + y_B)^{14-DCF}} \quad (2)$$

since there are 14 upcoming coupon payments.

In Excel, we can use the `YIELD(settlement, maturity, rate, pr, redemption, frequency, [basis])` function. `settlement` is again the date of valorization, i.e. 09/01/2026, `maturity` is the maturity date, set here to 22/04/2039, `rate` is the coupon rate, `pr` is the security's price per 100 face value, here 90.55, `redemption` is the amount repaid back to the bondholder at maturity, hence 100, `frequency` is the annual coupon frequency set to 1 and `basis` is set to 1 for "ACT/ACT" convention.

This function gives:

$$\begin{aligned} y_B &= \text{YIELD}(\text{DATE}(2026;1;9), \text{DATE}(2039;4;22), 2.75\%, 90.55, 100, 1, 1) \\ &= 3.660103\% \end{aligned}$$

One can check that Equation 2 is satisfied with this value of y_B .